JAMA Neurology | Original Investigation | AI IN NEUROLOGY

Detection of Epileptogenic Focal Cortical Dysplasia Using Graph Neural Networks
A MELD Study

Mathilde Ripart, PhD; Hannah Spitzer, PhD; Logan Z. J. Williams, MD; Lennart Walger, MSc; Andrew Chen, PhD; Antonio Napolitano, PhD; Camilla Rossi-Espagnet, MD, PhD; Stephen T. Foldes, PhD; Wenhan Hu, MD; Jiajie Mo, MD; Marcus Likeman, MD; Theodor Rüber, MD; Maria Eugenia Caligiuri, PhD; Antonio Gambardella, MD; Christopher Guttler, MD; Anna Tietze, MD, PhD; Matteo Lenge, PhD; Renzo Guerrini, MD; Nathan T. Cohen, MD; Irene Wang, PhD; Ane Kloster, MD; Lars H. Pinborg, MD; Khalid Hamandi, MD; Graeme Jackson, MD, PhD; Domenico Tortora, MD, PhD; Martin Tisdall, MD; Estefania Conde-Blanco, MD, PhD; Jose C. Pariente, MSc; Carmen Perez-Enriquez, PhD; Sofia Gonzalez-Ortiz, MD, PhD; Nandini Mullatti, MD; Katy Vecchiato, MD, PhD; Yawu Liu, MD, PhD; Reetta Kalviainen, MD, PhD; Drahoslav Sokol, PhD; Jay Shetty, MD; Benjamin Sinclair, PhD; Lucy Vivash, PhD; Anna Willard, MD, PhD; Gavin P. Winston, MD, PhD; Clarissa Yasuda, MD, PhD; Fernando Cendes, MD, PhD; Russell T. Shinohara, PhD; John S. Duncan, MD, PhD; J. Helen Cross, MD, PhD; Torsten Baldeweg, MD; Emma C. Robinson, PhD; Juan Eugenio Iglesias, PhD; Sophie Adler, MD, PhD; Konrad Wagstyl, MD, PhD; and the MELD FCD writing group

IMPORTANCE A leading cause of surgically remediable, drug-resistant focal epilepsy is focal cortical dysplasia (FCD). FCD is challenging to visualize and often considered magnetic resonance imaging (MRI) negative. Existing automated methods for FCD detection are limited by high numbers of false-positive predictions, hampering their clinical utility.

OBJECTIVE To evaluate the efficacy and interpretability of graph neural networks in automatically detecting FCD lesions on MRI scans.

DESIGN, SETTING, AND PARTICIPANTS In this multicenter diagnostic study, retrospective MRI data were collated from 23 epilepsy centers worldwide between 2018 and 2022, as part of the Multicenter Epilepsy Lesion Detection (MELD) Project, and analyzed in 2023. Data from 20 centers were split equally into training and testing cohorts, with data from 3 centers withheld for site-independent testing. A graph neural network (MELD Graph) was trained to identify FCD on surface-based features. Network performance was compared with an existing algorithm. Feature analysis, saliencies, and confidence scores were used to interpret network predictions. In total, 34 surface-based MRI features and manual lesion masks were collated from participants, 703 patients with FCD–related epilepsy and 482 controls, and 57 participants were excluded during MRI quality control.

MAIN OUTCOMES AND MEASURES Sensitivity, specificity, and positive predictive value (PPV) of automatically identified lesions.

RESULTS In the test dataset, the MELD Graph had a sensitivity of 81.6% in histopathologically confirmed patients seizure-free 1 year after surgery and 63.7% in MRI–negative patients with FCD. The PPV of putative lesions from the 260 patients in the test dataset (125 female [48%] and 135 male [52%]; mean age, 18.0 [IQR, 11.0-29.0] years) was 67% (70% sensitivity; 60% specificity), compared with 39% (67% sensitivity; 54% specificity) using an existing baseline algorithm. In the independent test cohort (116 patients; 62 female [53%] and 54 male [47%]; mean age, 22.5 [IQR, 13.5-27.5] years), the PPV was 76% (72% sensitivity; 56% specificity), compared with 46% (77% sensitivity; 47% specificity) using the baseline algorithm. Interpretable reports characterize lesion location, size, confidence, and salient features.

CONCLUSIONS AND RELEVANCE In this study, the MELD Graph represented a state-of-the-art, openly available, and interpretable tool for FCD detection on MRI scans, with significant improvements in PPV. Its clinical implementation holds promise for early diagnosis and improved management of focal epilepsy, potentially leading to better patient outcomes.

JAMA Neurol. 2025;82(4):397-406. doi:10.1001/jamaneurol.2024.5406
Published online February 24, 2025. Last corrected on April 14, 2025.

Supplemental content

Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: The MELD FCD writing group authors are listed at the end of the article.

Corresponding Author: Mathilde Ripart, PhD, UCL Great Ormond Street Institute of Child Health, University College London, 30 Guilford St, London WC1N 1EH, United Kingdom (m.ripart@ucl.ac.uk).

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

Focal cortical dysplasia (FCD) is a leading cause of drug-resistant focal epilepsy in children and adults.[1] It is surgically remediable with postoperative seizure freedom rates of around 65%,[2] but lesions are often considered magnetic resonance imaging (MRI) negative and postsurgical outcome is highly affected by early and accurate detection of lesions on MRI.[3,4]

Machine learning methods to detect FCD on MRI have improved in recent years.[5-7] However, clinical utility of these approaches is hampered by the multiple putative lesions identified, resulting in a low positive predictive value (PPV) for each being a true FCD.[8] Previous approaches have addressed constraints on sample sizes and computational memory by subdividing MRI scans into smaller independent patches.[5-7] This limits their capacity to prioritize across disparate putative abnormalities.

Here, through the Multicenter Epilepsy Lesion Detection (MELD) Project, we leveraged advances in deep learning[9-14] to train a graph neural network to segment FCD on a large multicenter MRI cohort of patients (**Figure 1**). The novel MELD Graph algorithm, which incorporates whole-brain context to improve specificity and accuracy, was directly compared against our previously published patch-based multilayer perceptron,[5] including on a multicenter independent test cohort. We aimed to create a state-of-the-art, clinically translatable, open-access artificial intelligence (AI) algorithm for the automated detection of FCDs.

Key Points

Question Can the diagnosis of epilepsy-causing focal cortical dysplasias (FCD) be improved using state-of-the-art artificial intelligence?

Findings In this diagnostic study of 703 patients with epilepsy due to FCD, a context-aware graph neural network (MELD Graph) detected 64% of lesions previously missed by radiologists with a high positive predictive value. Interpretable reports may facilitate clinical integration by characterizing lesion location, size, and morphology alongside the algorithm's prediction confidence.

Meaning This publicly available interpretable algorithm, validated on a large cohort, can advance the use of AI-based radiological adjuncts for early detection and neurosurgical planning in patients with focal epilepsy due to FCD.

Methods

Dataset

The MELD Project used an MRI dataset of 1185 participants collated from 23 international epilepsy surgery centers (Figure 1; eTable 1 in Supplement 1). Each center received local ethical approval to retrieve and anonymize retrospective routinely available clinical data without requiring explicit consent. Standards for Reporting of Diagnostic Accuracy (STARD) reporting guidelines were followed.

Patients were included if they had a radiological or histopathological diagnosis of FCD. Controls were included if they were scanned for research purposes or for headache and if they had no other neurological conditions and normal MRI results. For full details on demographic and clinical data and MRI data processing, see the eMethods in Supplement 1.

Three-dimensional (3-D) T1-weighted (all participants) and T2-weighted fluid-attenuated inversion recovery (FLAIR) (where available) MRI images were processed using FreeSurfer[15] and 11 features were extracted: cortical thickness, gray-white matter intensity contrast, intrinsic curvature, sulcal depth, curvature, and FLAIR intensity sampled at 6 intracortical and subcortical depths.

3-D regions of interest (ROIs) for each FCD were manually delineated on the MRI.[16] If there was no visible lesion (MRI-negative), resection cavities on postsurgical scans were used to guide masking. ROIs were intersected with FreeSurfer surfaces and the features and ROIs were registered to a bilaterally symmetrical template[17] (Figure 1A). Quality control involved outlier identification and removal of patients with missing lesion masks.[18]

The main cohort, comprising data from 20 centers, was randomly split 50:50 into training and testing cohorts (**Table 1**), matching the randomization of a previous study[5] to enable direct comparison. Each center had participants in both training and testing cohorts. An additional independent test cohort included 3 additional centers.

Surface-based features were preprocessed into smoothed features, control-normalized features, and feature asymmetries (eMethods in Supplement 1). An additional feature was cortical thickness with curvature regressed out.[19] This resulted in 34 input features per participant calculated at 163 842 cortical surface vertices. MRI features vary depending on the scanner, worsening algorithm performance on untested scanners. To adjust for scanner effects, extracted features in the independent test cohorts underwent intersite harmonization to the training cohort using distributed ComBat.[20] To estimate the number of participants needed to generate consistent harmonization parameters, a series of subsampling experiments was performed (eMethods in Supplement 1). Model performance on the independent test cohorts was evaluated with and without harmonization.

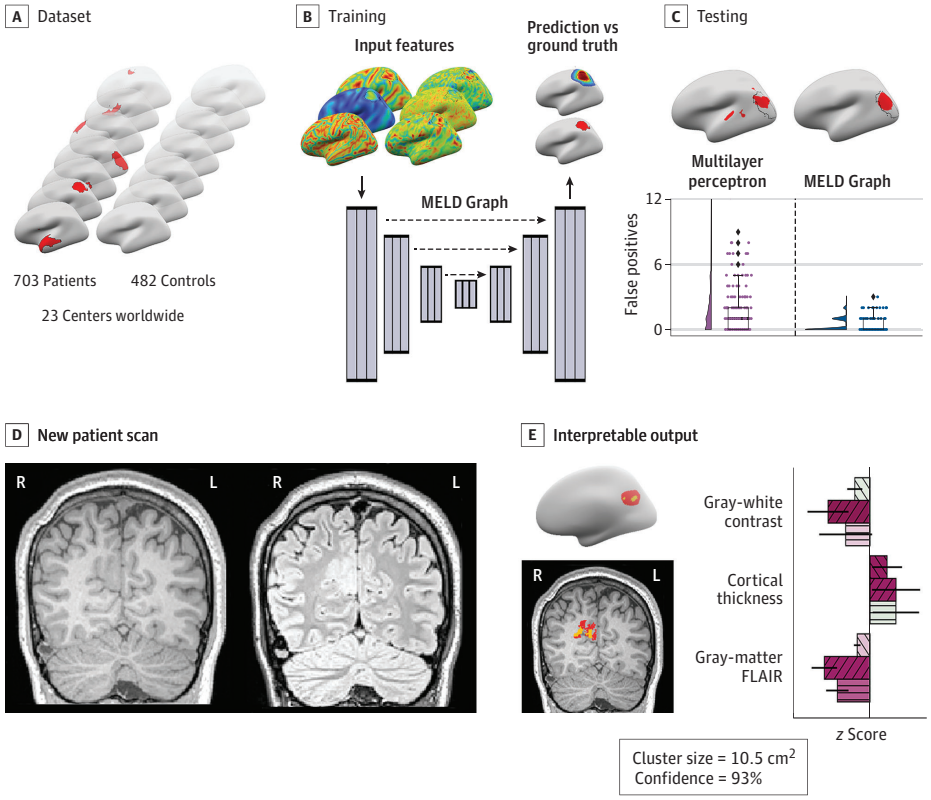
MELD Graph for FCD Lesion Segmentation

Previous multicenter-validated FCD detection tools restricted model inputs to small 1-cm^3 patches of cortex. Deep-learning models, which have the capacity to extract increasingly abstract, complex, and large-scale features, have revolutionized biomedical image segmentation. nnU-Net,[13] a convolutional neural network, achieves state-of-the-art performance across a range of biomedical segmentation tasks but is typically applied to regular 2-dimensional/3-D grids with translation-invariant convolutions. To address the irregularity of folded cortical surfaces, we developed MELD Graph, a graph-based nnU-Net implementation (Figure 1B; eFigure 1 and eMethods in Supplement 1).[11,13,21] MELD Graph processes hemispheres with 34 features and 163 842 vertices, predicting lesion segmentations with neighbor-connected lesional vertices grouped into single clusters.

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

Figure 1. Overview of Multicenter Epilepsy Lesion Detection (MELD) Graph



A, The MELD focal cortical dysplasia (FCD) dataset contains surface-based magnetic resonance imaging data and ground truth lesion masks for 703 patients and 482 controls from 23 centers worldwide. B, The MELD Graph U-Net model was trained to recognize patterns of abnormal features and their broader hemispheric context to segment FCD lesions. C, The model was compared against a previously published and widely reused baseline algorithm—a multilayer perceptron.[5] MELD Graph had a significantly increased positive predictive value driven by a decrease in the number of false-positive clusters. D, Images from a new patient with a suspected FCD can be analyzed with MELD Graph to generate an interpretable report (E) that displays predicted lesion location, size, and features, alongside confidence and relative feature importance (salience). L indicates left; R, right; FLAIR, fluid-attenuated inversion recovery.

Evaluation of the MELD Graph Model

The MELD Graph model and a baseline, previously published MELD Multi Layer Perceptron (MELD MLP) model[5] were evaluated on the test and independent test cohorts (Figure 1C). The following metrics were calculated: sensitivity, specificity, PPV, and Intersection over Union (IoU)—a metric for segmentation accuracy (eMethods in Supplement 1). MELD Graph lesion detection rates, where a predicted lesion cluster overlapped with (or was within 20 mm of) the manual lesion mask, in the test dataset were stratified by demographic factors (eMethods in Supplement 1). Confidence intervals and significant differences between models and demographic factors ($P < .05$) were calculated using bootstrapping and permutation-based null models (eMethods in Supplement 1). As an additional measure of model specificity, the number of false-positive clusters (defined as predictions beyond 2 cm of a lesion mask) in patients and controls was calculated.

MELD Graph performances were compared with a similar model but trained solely on a subset of patients with MRI-negative FCD confirmed histopathologically and controls to establish whether selective training improved performance on these patients compared with the larger, heterogeneous dataset (eMethods in Supplement 1).

Table 1. Comparison of the Performance of the Multicenter Epilepsy Lesion Detection (MELD) Graph Model to the Original Multilayer Perceptron (MLP) on the Same Withheld and Independent Test Cohort Derived from 3 Centers

	Median (95% CI)	
Model	MELD MLP	MELD graph
Test dataset		
Sensitivity, % (n = 260 patients)	67 (61-73)	70 (64-75)
Specificity, % (n = 193 controls)	54 (47-61)	60 (53-67)
PPV, %	39 (35-44)	67 (62-73)[a]
IoU (n = 160)	0.23 (0.22-0.26)	0.3 (0.27-0.33)[a]
Independent test cohort		
Sensitivity, % (n = 116 patients)	77 (69-84)	72 (61-78)
Specificity, % (n = 101 controls)	47 (37-56)	56 (47-66)[a]
PPV, %	46 (40-53)	76 (61-79)[a]
IoU (n = 78)	0.29 (0.25-0.33)	0.36 (0.31-0.41)[a]

Abbreviations: IoU, intersection over union; PPV, positive predictive value.

[a] Metrics with significantly changed performance ($P < .05$) in comparison with MELD MLP.

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

Interpretable AI

For all predicted lesional vertices, integrated gradients saliency was computed,[22] identifying the relative importance of input variables for a given prediction. Saliency can be averaged across input features to identify the most salient vertices within a lesion or averaged across lesional vertices to identify which input features were most important to the model's prediction (eMethods in Supplement 1). To highlight salient regions, the 20% most salient lesional vertices of each predicted lesion were identified. For these salient vertices, we computed the prediction confidence (maximum model prediction score), mean lesional feature values, and mean feature salience. The calibration of confidence scores was assessed using expected calibration error.

To characterize the true-positive and false-positive predictions, as well as the missed FCD lesions (false negatives), mean feature values for these regions were calculated. Detected FCDs were characterized according to their histological subtypes. Additionally, we characterized model performance (sensitivity and PPV) according to lesion size and lesion location (eMethods in Supplement 1).

Individual Reports

The trained model was incorporated into a pipeline that takes input MRI data (Figure 1D; eFigure 2 in Supplement 1), performs feature extraction and preprocessing, runs inference on the MELD Graph model, and generates an interpretable report (Figure 1E; eFigure 2 in Supplement 1). The report details the locations of predicted lesions (and the 20% most salient vertices) on the cortical surface and native T1, as well as model confidence in lesional predictions, lesional features, and the associated saliencies of features.

Code and Data Availability

All code is available to download.[23] Lesion masks in template space are available.[24] Access to the MELD surface-based FCD dataset is via request.

Results

Participants

There were 57 participants removed during quality control (37 patients with missing lesion masks, 13 outlier participants, and 7 participants where FreeSurfer failed). Data from 20 centers were split into training (278 patients; 180 controls) and testing (260 patients; 193 controls) cohorts (eTable 1 in Supplement 1). Participants from 3 centers were withheld as an independent test cohort (116 patients; 101 controls). Lesion masks for patients previously considered MRI-negative were significantly smaller (median, 29% smaller) than visible lesions (Mann-Whitney U, $z = -2.07$; $P < .04$).

Model Evaluation

The performance of the MELD Graph model compared with the baseline multilayer perceptron (MELD MLP) on the test dataset is presented in Table 1. MELD Graph had significantly higher PPV than the baseline MELD MLP model on test (67% vs 39%) and independent (76% vs 46%) test cohorts, primarily due to a reduction in the number of false-positive clusters (**Figure 2**). On the test dataset, MELD Graph had a maximum of 3 false-positive clusters in patients (median, 0; IQR, 0-1) and 2 in controls (median, 0; IQR, 0-1), in comparison with MELD MLP for which patients had a maximum of 12 clusters (median, 1; IQR, 0-2), and 8 in controls (median, 0; IQR, 0-1) (eTable 2 in Supplement 1). This significant reduction in false positives for MELD Graph was also seen on the independent test cohort (eTable 2 in Supplement 1). In patients whose lesions were detected by both the MELD Graph and MELD MLP models, the segmentation accuracy, computed as IoU, was significantly higher using MELD Graph in the test (0.3 vs 0.23 [N = 160]) and independent test (0.36 vs 0.29 [N = 78]) cohorts (Table 1). Figure 2 provides examples of individual predictions using the MELD Graph and MLP models. MELD Graph predictions do not have additional predictions (false positives) and segmentations are smoother and more contiguous.

Performance on the independent test cohort was calculated with and without harmonization (eTable 3 in Supplement 1). Sensitivity remained stable, 72% with and 70% without harmonization, whereas specificity dropped from 56% with to 39% without harmonization. Subsampling experiments indicated that a minimum of 20 participants was required to generate reliable harmonization parameters (eFigure 3 in Supplement 1). Moreover, training MELD Graph solely on MRI-negative, histopathologically confirmed FCDs showed a significant drop in the PPV, from 72% to 58% (eTable 4 in Supplement 1).

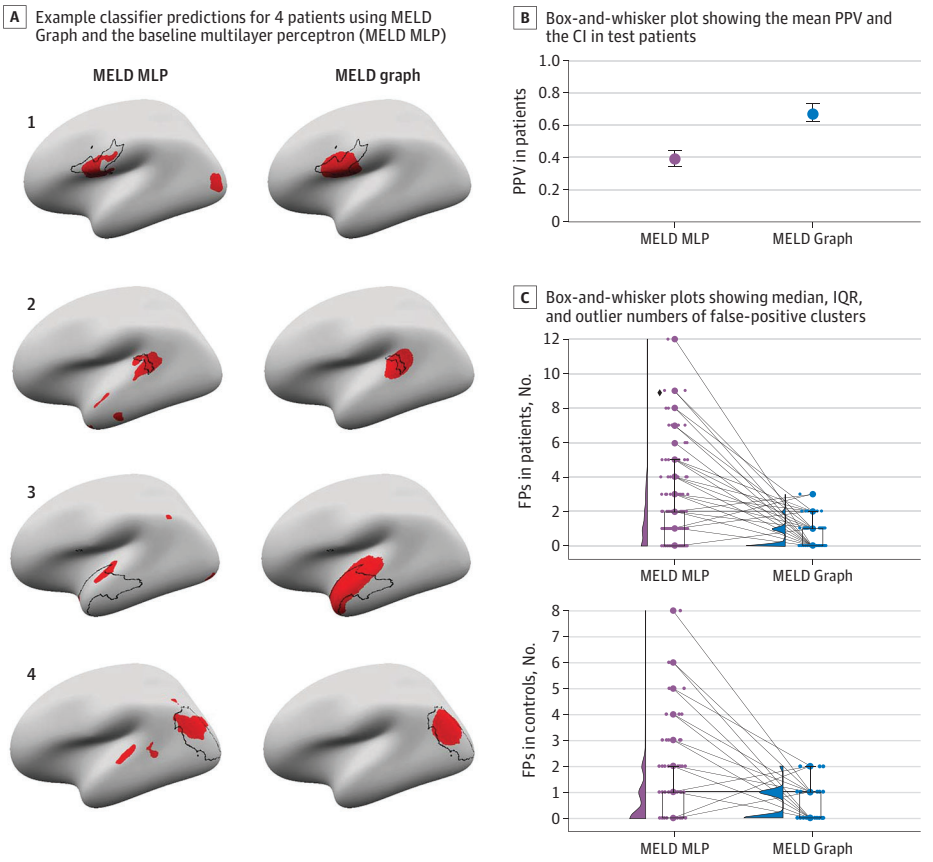
Table 2 provides a performance breakdown according to demographic factors. There was a 63.7% detection rate (51 of 80) in patients previously reported MRI-negative. MELD Graph detected 75.4% of 57 type IIA and 76.3% of the 93 type IIB lesions. In seizure-free patients with histopathologically confirmed FCDs, 81.6% of lesions were detected. Of note, MELD Graph was able to detect 84.6% (11 of 13) of the particularly subtle FCD type I lesions. Histopathological subgroups had no significant impact on sensitivity. However, sensitivity was lower in the group of patients without histopathological confirmation, either because they had not been operated on or because histopathology was inconclusive. Lower detection rates in this group might reflect uncertainty in their manually defined masks.

Analysis of detected and missed lesions in the test cohort (eFigure 4 in Supplement 1) revealed features driving the algorithm's prediction of lesional vertices. Overall, lesions were characterized by abnormally deep sulci, increased intrinsic curvature and cortical thickness, decreased gray-white matter contrast, decreased gray matter FLAIR intensity, and increased white matter FLAIR intensity. FCD types IIA and IIB demonstrated pronounced folding and thickness abnormalities, with FCD type IIB having the additional distinctive FLAIR hyperintensity in the white matter (transmantle sign). In comparison, FCD types I and III had more subtle cortical tissue abnormalities. Lesions not detected by MELD Graph were characterized by having less abnormal features (eFigure 4 in Supplement 1). Model performance was not significantly associated with either lesion size or location (eFigure 4 in Supplement 1). In detected clusters in the test cohort, IoU correlated

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

Figure 2. Reduction in False-Positive Clusters With Multicenter Epilepsy Lesion Detection (MELD) Graph



A, Example classifier predictions for 4 patients using MELD Graph and the baseline multilayer perceptron (MELD MLP). The black line indicates manual lesion mask and the red line indicates classifier predictions. B, Box-and-whisker plot showing the mean positive predictive value (PPV) and the confidence interval (CI) in the test patients detected by both MELD MLP and MELD Graph models (N = 161). C, Box-and-whisker plot showing median, IQR, and outlier numbers of false positive (FPs) clusters predicted on patients and controls in the test dataset using MELD Graph compared with MELD MLP. Gray lines connect identical subjects between the models.

significantly with the confidence score (r = 0.55; P < .01), indicating higher segmentation accuracies in higher confidence predictions (eFigure 5 in Supplement 1).

Interpretable AI Models

The MELD Graph tool outputs model predictions in native space on the T1-weighted scan in Neuroimaging Informatics Technology Initiative (NIfTI) format and an interpretable report in PDF format, containing predicted lesion locations and associated model confidence, as well as characterizing the lesional features and their salience for each putative lesion. **Figure 3** and eFigure 6 in Supplement 1 contain example components from individual patient reports. Patient 1 has a lesion in the precuneus. MELD Graph was 93% confident in its prediction. The lesion, as evident by the feature z scores, is characterized by blurring of the gray-white matter boundary (decreased gray-white contrast), increased cortical thickness, and abnormal FLAIR signal intensity in the gray matter. Most of these features have high saliency scores indicating their importance to the classifier's prediction. Patients 2 through 4 are examples of MRI-negative lesions not identified by 5 expert raters.[25] MELD Graph identified these lesions with 7% to 45% confidence. Interpretable reports for these patients are available.[23]

The expected calibration error, which quantifies the discrepancy between MELD Graph's confidence scores and the observed accuracies, was calculated as 0.10 (where 0 represents perfect calibration). This indicates that model confidence scores were well calibrated (eFigure 5 in Supplement 1).

Discussion

MELD Graph is a graph convolutional neural network for automated segmentation of FCD that integrates information across whole cortical hemispheres. The model was tested on a large dataset of 453 participants (from 20 centers), as well as an independent test cohort of 217 participants from 3 additional centers. MELD Graph accurately localized 81.6% of histopathologically confirmed FCDs in patients seizure-free 1 year after surgery. Over the entire dataset, MELD Graph accurately localized 70% of the FCDs from the test dataset and 72% from the independent test cohort, with a 4-fold reduction in the number of false-positive clusters and a significantly improved PPV over previous approaches. MELD Graph is released as an open-access tool that takes input MRI data (a preoperative T1-weighted scan plus an optional FLAIR scan) from a patient with epilepsy with a suspected FCD, performs feature extraction and processing, runs inference using the MELD Graph model, and generates an interpretable report depicting lesion location, model confidence, and characterizing lesional features and their salience.

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

Table 2. Multicenter Epilepsy Lesion Detection Graph Performance on the Test Dataset, Grouped According to Demographic Factors

Characteristic	% Detected[a]	Patients, No.
Age group		
Adult	67.9	131
Pediatric	71.3	129
Sex		
Female	62.4	125
Male	76.3[b]	135
Ever reported MRI-negative		
Visible	72.2	180
MRI-negative	63.7	80
Postoperative seizure freedom		
Seizure free	79.2	106
Not seizure free	62.7[b]	51
Histology		
FCD I	84.6	13
FCD IIA	75.4	57
FCD IIB	76.3	93
FCD III	75.0	8
Not available	56.2[b]	89
Histology confirmed and seizure free		
True	81.6[b]	98
False	62.3	162
Modality		
T1 weighted only	67.3	150
T1 weighted and FLAIR	72.7	110

Abbreviations: FCD, focal cortical dysplasia; MRI, magnetic resonance imaging.

[a] FCD detection rate is broken down by age group, sex, MRI status, seizure freedom, histology, and available MRI modalities. Lesions termed *MRI-negative* were from MRI scans reported as nonlesional at some point during the patient's clinical evaluation.

[b] $P < .05$.

Automated Tools to Detect FCDs

FCD is the most common histopathological finding in surgical patients with MRI-negative epilepsy.[26,27] In MRI-negative patients, the absence of a confident lesional hypothesis can prohibit surgical candidacy and additional noninvasive[28] and invasive[29] investigations to identify a lesion can introduce significant delays to surgery. Increased duration of epilepsy is associated with poorer outcome and progressive cognitive sequelae.[2,4] Surgery is curative in around 65% and seizure freedom combined with reductions in antiseizure medications may halt or even reverse neuropsychological decline.[30] The development of tools, like MELD Graph, that can successfully localize FCDs across a range of scanners, ages, and subtypes, including detecting 64% of MRI-negative FCDs, therefore, may improve both seizure freedom and developmental outcomes[30] through earlier diagnosis and surgery.[31]

Previous models for detecting FCD have been validated on multicenter data,[5-7] but have a recognized propensity to produce multiple false-positive predictions in patients,[8] hampering their utility as radiological adjuncts. These models have analyzed limited portions of the brain in isolation, either as small patches of surface vertices[18,32,33] or voxels.[6,7] In contrast, the MELD Graph model incorporates whole-brain context, leading to multiple improvements: increased specificity (Figure 2), heightened sensitivity, especially to subtle FCD type I lesions (Table 2), better accuracy (Table 1), and well-calibrated confidence estimates (eFigure 5 in Supplement 1). Evaluated on the same dataset, MELD Graph produced a maximum of 3 clusters in patients, whereas the MELD MLP model identified up to 12. For clinical translation, this will reduce the time necessary for clinicians to review model outputs and improve confidence in AI radiological adjuncts. Future integration of MELD Graph into clinical practice may lead to earlier diagnosis of some patients, as well as the detection of subtle lesions in some patients currently considered nonlesional. MELD Graph outputs could be considered in conjunction with other investigations, such as electroencephalography, magnetoencephalography, and positron emission tomography, or used in the planning of stereoelectroencephalography.[34,35] This may expedite patients' journeys to epilepsy surgery, increasing the likelihood of seizure freedom,[2] as well as cessation of neurocognitive decline.[30]

Interpretable Models for Clinical Translation

Model interpretability is critical for incorporating AI into radiological review. MELD Graph reports (Figure 3 and eFigure 6 in Supplement 1) highlight lesion location, imaging characteristics, the lesional patches that were most salient to the model,[22] and the model's confidence in its prediction.[36] High-confidence predictions are rarely incorrect, while lower-confidence outputs may highlight subtle abnormalities that warrant careful review (eFigure 5 in Supplement 1).

MRI identification and delineation of FCD is often carried out at multiple stages during a patient's journey and the development of lesion detection tools must address these use cases. These include early review of an MRI by a general radiologist at a nonspecialist center, expert neuroradiological review of a patient referred to a surgical center,[37] multidisciplinary review at an epilepsy surgery meeting,[38] neurosurgical planning of stereoelectroencephalography implantation,[34,35] and neurosurgical planning of lesion resection or ablation.[39] By providing confidence estimates for each putative lesion, clinicians and researchers using MELD Graph can individually choose a confidence threshold, balancing their need for sensitivity vs specificity. For example, a neuroradiologist at a specialist epilepsy surgery center may choose to review low-confidence clusters in a complicated MRI-negative patient. The interpretable lesion reports, alongside confidence scores, are a step toward clear communication of model outputs to facilitate translation.

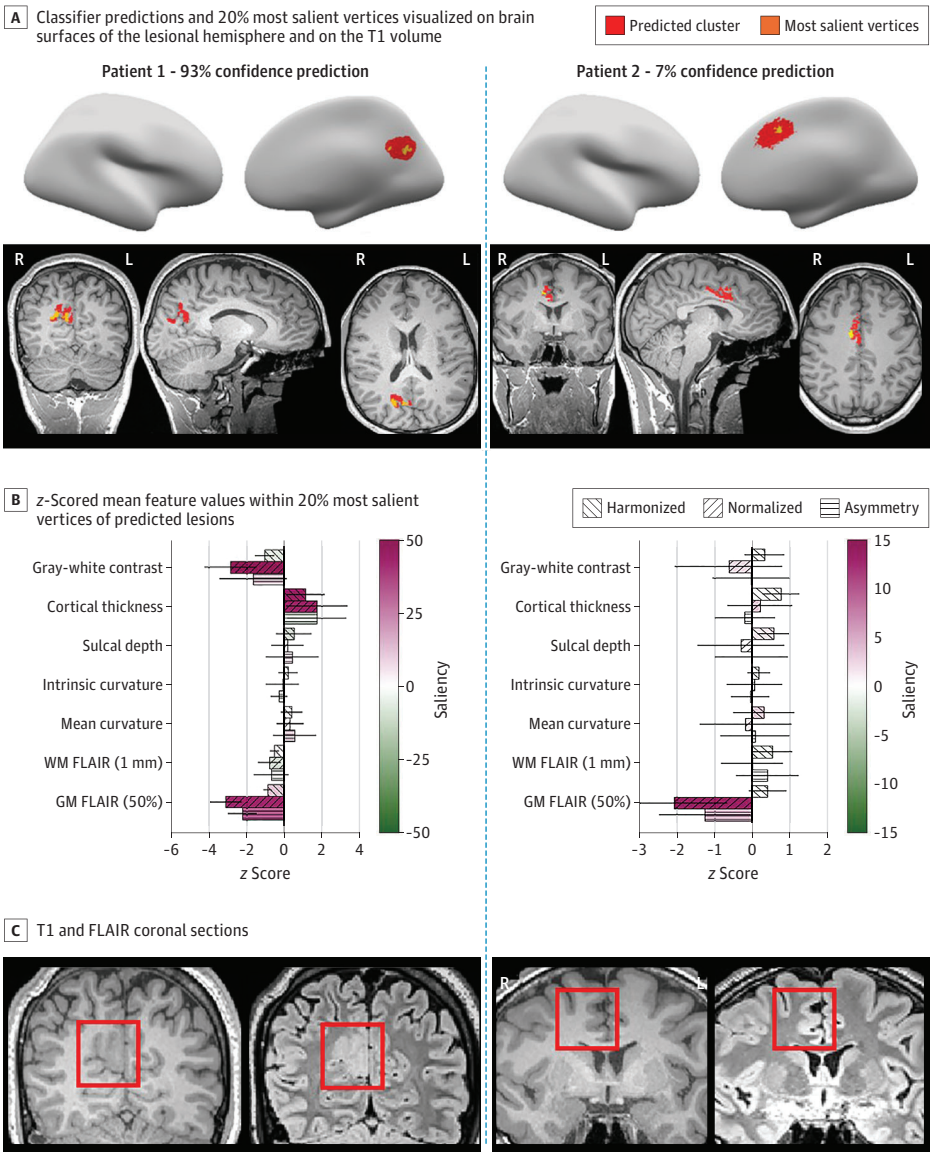
Validation of Neurotechnology

One challenge for many promising AI algorithms is that their performance typically drops on data from new patients in new settings. Previously unseen MRI scanners, with differing sequence parameters resulting in subtly different image contrasts, represent a domain shift from the training data distribution. Using multicenter data during training is one mitigation strategy, as the algorithm is exposed to some of this hetero-

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

Figure 3. Examples of Interpretable Patient Reports



Multicenter Epilepsy Lesion Detection (MELD) Graph outputs for 2 patients from the independent test cohort. Patient 1 is an example of an MRI-positive focal cortical dysplasia (FCD) detected by MELD Graph with a high-confidence prediction (93%). Patient 2 has an FCD that was not identified by 5 expert radiologists but detected by MELD Graph with a low-confidence prediction (7%). A, Classifier predictions and 20% most salient vertices visualized on brain surfaces of the lesional hemisphere and on the T1 volume. B, *z*-scored mean feature values within 20% most salient vertices of predicted lesions. Color represents saliency scores. Features driving the classifier's prediction have positive saliency and features inconsistent with MELD Graph prediction have negative saliency. C, T1 and fluid-attenuated inversion recovery (FLAIR) coronal sections with a red box indicating the lesional cortex. L indicates left; R, right.

geneity. However, an essential validation of new machine-learning technologies is testing them on data from withheld, independent test sites. Here, we demonstrated that MELD Graph maintained performance on unseen data from 3 centers. Optimal specificity on independent test site data was dependent on harmonization of new data to the training cohort. However, this harmonization process requires a minimum of 20 scans acquired on the same scanner (eFigure 3 in Supplement 1). Therefore, to facilitate reuse of the algorithm on individual MRI scans from patients from new centers, for which such harmonization is not possible, a nonharmonization version is provided with documented performance statistics (eTable 3 in Supplement 1). Furthermore, the release of MELD Graph as an open-source tool serves to enable independent validation and reuse, including prospective studies to assess whether it accelerates diagnosis, alters treatment, and ultimately improves outcomes.

Limitations

Surface-based features have a number of advantages, including interpretability and the ability to share anonymized data for reanalysis.[40,41] Nevertheless, the use of predetermined features limits the potential for the network to learn novel features. Future work, applying deep learning to multimodal data, including MRI, positron emission tomography, magnetoencephalography, and electroencephalography, will enable complex features to be learned that may not be visible or known to neuroradiologists. Lastly, MELD Graph has not been evaluated on patients with multiple FCDs.

Conclusions

MELD Graph is a state-of-the-art, openly available, graph convolutional network for FCD detection. With improved perfor-

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

mance over existing methods, interpretable predictions, model confidence scores, and individual patient reports, MELD Graph can support the integration of lesion detection tools into the radiological workflow.

ARTICLE INFORMATION

Accepted for Publication: October 29, 2024.

Published Online: February 24, 2025. doi:10.1001/jamaneurol.2024.5406

Correction: This article was corrected on April 14, 2025, to fix typographical errors in the Methods and Results section, to correct Figure 2, and to update the Conflict of Interest Disclosures.

Author Affiliations: UCL Great Ormond Street Institute of Child Health, London, United Kingdom (Ripart, Cross, Baldeweg, Adler, Wagstyl); Institute for Stroke and Dementia Research, University Hospital, LMU Munich, Germany (Spitzer); Institute of Computational Biology, Helmholtz Munich, Germany (Spitzer); School of Biomedical Engineering & Imaging Sciences, King's College London, London, United Kingdom (Williams, Robinson, Wagstyl); Department of Neuroradiology, University Hospital Bonn, Bonn, Germany (Walger, Rüber); Department of Epileptology, University Hospital Bonn, Bonn, Germany (Walger, Rüber); Penn Statistics in Imaging and Visualization Center, Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania (Chen, Shinohara); Medical Physics Unit, Bambino Gesù Children's hospital, IRCCS, Rome, Italy (Napolitano); Functional and Interventional Neuroimaging Unit, Bambino Gesù Children's hospital, IRCCS, Rome, Italy (Rossi-Espagnet); Barrow Neurological Institute, Phoenix, Arizona (Foldes); Beijing Tiantan Hospital, Beijing, China (Hu, Mo); Bristol Royal Hospital for Children, Bristol, United Kingdom (Likeman); Department of Medical and Surgical Sciences, University Magna Graecia, Catanzaro, Italy (Caligiuri, Gambardella); Charité—Universitätsmedizin Berlin, Germany (Guttler, Tietze); Freie Universität Berlin, Germany (Guttler, Tietze); Institute of Neuroradiology, Humboldt-Universität zu Berlin, Berlin, Germany (Guttler, Tietze); Neuroscience and Human Genetics Department, Meyer Children's Hospital IRCCS, Florence, Italy (Lenge, Guerrini); University of Florence, Florence, Italy (Lenge, Guerrini); Center for Neuroscience, Children's National Hospital, Washington, DC (Cohen); Epilepsy Center, Neurological Institute, Cleveland Clinic, Cleveland, Ohio (Wang); Epilepsy Clinic & Neurobiology Research Unit, Department of Neurology, Copenhagen University Hospital—Rigshospitalet, Copenhagen, Denmark (Kloster, Pinborg); University Hospital of Wales, Cardiff, United Kingdom (Hamandi); The Florey Institute of Neuroscience and Mental Health, University of Melbourne, Australia (Jackson); Comprehensive Epilepsy Program, Austin Health, University of Melbourne, Victoria, Australia (Jackson); Department of Neuroradiology, IRCCS Istituto Giannina Gaslini, Genova, Italy (Tortora); Great Ormond Street Hospital for Children, London, United Kingdom (Tisdall, Vecchiato); Department of Neuroradiology, Hospital Clinic & Fundació de Recerca Clínic Barcelona-Institut d'Investigacions Biomèdiques August Pi i Sunyer, Barcelona, Spain (Pariente); Member of EpiCARE ERN (Kalviainen, Adler); Epilepsy Monitoring Unit, Department of Neurology, Hospital del Mar, Barcelona, Spain (Perez-Enriquez); Epilepsy Unit, Department of Neurology, Hospital Vithas Málaga, Spain (Perez-Enriquez); Department of Neuroradiology, Hospital Clinic, Barcelona, Spain (Gonzalez-Ortiz); Department of Clinical Neurophysiology and Epilepsy, Kings College Hospital, London, United Kingdom (Mullatti); Department of Neurology, University of Eastern Finland, Kuopio, Finland (Liu, Kalviainen); Kuopio Epilepsy Center, Kuopio University Hospital, Kuopio, Finland (Kalviainen); Paediatric Neurosciences, Royal Hospital for Children and Young People, Edinburgh, United Kingdom (Sokol, Shetty); Department of Neuroscience, School of Translational Medicine, Alfred Health and Monash University, Melbourne, Australia (Sinclair, Vivash, Willard); UCL Queen Square Institute of Neurology, London, United Kingdom (Winston, Duncan); Department of Medicine, Queen's University, Kingston, Canada (Winston); UNICAMP University of Campinas, Campinas, Brazil (Yasuda, Cendes); Brazilian Institute of Neuroscience and Neurotechnology, Campinas, Brazil (Yasuda, Cendes); National Hospital for Neurology and Neurosurgery, London, United Kingdom (Duncan); Massachusetts General Hospital & Harvard Medical School, Boston, Massachusetts (Iglesias); Centre for Medical Image Computing, UCL, United Kingdom (Iglesias); Department of Neurology, Hospital Clínic & Fundació de Recerca Clínic Barcelona-Institut d'Investigacions Biomèdiques August Pi i Sunyer, Barcelona, Spain (Conde-Blanco).

The MELD FCD Writing Group Authors: Abdulah Fawaz, PhD; Alessandro De Benedictis, MD, PhD; Luca De Palma, MD; Kai Zhang, MD; Angelo Labate, MD; Carmen Barba, MD, PhD; Xiaozhen You, PhD; William D. Gaillard, MD; Yingying Tang, MD; Shan Wang, MD; Shirin Davies, MSc; Mira Semmelroch, MSc; Mariasavina Severino, MD; Pasquale Striano, MD, PhD; Aswin Chari, MD, PhD; Felice D'Arco, MD; Kshitij Mankad, MD; Nuria Bargallo, MD, PhD; Saul Pascual-Diaz, PhD; Ignacio Delgado-Martinez, MD, PhD; Jonathan O'Muircheartaigh, PhD; Eugenio Abela, MD; Jothy Kandasamy, MD; Ailsa McLellan, MD; Patricia Desmond, MD; Elaine Lui, MD, PhD; Terence J. O'Brien, MD; Kirstie Whitaker, PhD.

Affiliations of The MELD FCD Writing Group Authors: School of Biomedical Engineering & Imaging Sciences, King's College London, London, United Kingdom (Fawaz); Neurosurgery Unit, Bambino Gesù Children's Hospital, IRCCS, Rome, Italy (De Benedictis); Neurology Epilepsy and Movement Disorders, Bambino Gesù Children's Hospital, IRCCS, Rome, Italy (De Palma); Beijing Tiantan Hospital, Beijing, China (Zhang); University of Messina, Italy (Labate); Neuroscience and Human Genetics Department, Meyer Children's Hospital IRCCS, Florence, Italy (Barba); University of Florence, Florence, Italy (Barba); Center for Neuroscience, Children's National Hospital, Washington, DC (You, Gaillard); Epilepsy Center, Neurological Institute, Cleveland Clinic, Cleveland, Ohio (Tang, Wang); Department of Neurology, West China Hospital of Sichuan University, Chengdu, Sichuan, China (Tang); Epilepsy Center, Department of Neurology, Second Affiliated Hospital, School of Medicine, Zhejiang University, Hangzhou, Zhejiang, China (Wang); University Hospital of Wales, Cardiff, United Kingdom (Davies); The Florey Institute of Neuroscience and Mental Health, University of Melbourne, Australia (Semmelroch); Department of Neuroradiology, IRCCS Istituto Giannina Gaslini, Genova, Italy (Severino); Department of Neurosciences, Rehabilitation, Ophthalmology, Genetics, Maternal and Child Health, Genova, Italy (Striano); Great Ormond Street Hospital for Children, London, United Kingdom (Chari, D'Arco, Mankad); Department of Neuroradiology, Hospital Clinic & Fundació de Recerca Clínic Barcelona-Institut d'Investigacions Biomèdiques August Pi i Sunyer, Barcelona, Spain (Bargallo); Member of EpiCARE ERN (Striano, Bargallo); Fundació de Recerca Clínic Barcelona-Institut d'Investigacions Biomèdiques August Pi i Sunyer, Barcelona, Spain (Pascual-Diaz); Human Anatomy and Embryology Unit, Department of Morphological Sciences, Faculty of Medicine, Universitat Autònoma de Barcelona, Bellaterra, Spain (Delgado-Martinez); Department of Neurosurgery, Hospital del Mar, Barcelona, Spain (Delgado-Martinez); Zentrum für Entwicklungs- und Neuropsychiatrie, Klinik für Konsiliar, Alters- und Neuropsychiatrie, Psychiatrische Dienste Argau AG, Windisch, Switzerland (Abela); Forensic and Neurodevelopmental Sciences, Institute of Psychiatry, King's College London, United Kingdom (O'Muircheartaigh); Department of Clinical Neurosciences, Royal Hospital for Children and Young People, Edinburgh, United Kingdom (Kandasamy); Paediatric Neurosciences, Royal Hospital for Children and Young People, Edinburgh, United Kingdom (McLellan); Department of Radiology, Royal Melbourne Hospital, University of Melbourne, Australia (Desmond, Lui); Department of Neuroscience, School of Translational Medicine, Alfred Health and Monash University, Melbourne, Australia (O'Brien); Department of Medicine, The Royal Melbourne Hospital, The University of Melbourne, Victoria, Australia (O'Brien); The Alan Turing Institute, London, United Kingdom (Whitaker).

Author Contributions: Dr Ripart had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Drs Adler and Wagstyl contributed equally as joint-senior authors.
Concept and design: Ripart, Spitzer, Fawaz, Chen, De Benedictis, Whitaker, Cross, Baldeweg, Robinson, Iglesias, Adler, Wagstyl.
Acquisition, analysis, or interpretation of data: Ripart, Spitzer, Williams, Walger, De Palma, Napolitano, Rossi-Espagnet, Foldes, Zhang, Hu, Mo, Likeman, Rüber, Caligiuri, Gambardella, Labate, Güttler, Tietze, Barba, Lenge, Guerrini, You, Cohen, Gaillard, Tang, S. Wang, I. Wang, Kloster, Pinborg, Davies, Hamandi, Semmelroch, Jackson, Tortora, Severino, Striano, Chari, D'arco, Mankad, Tisdall, Bargalló, Conde-Blanco, Pascual-Diaz, Pariente, Delgado-Martinez, Pérez-Enríquez, González-Ortiz, Abela, Mullatti, Vecchiato, O'Muircheartaigh, Liu, Kalviainen, Kandasamy, McLellan, Sokol, Shetty, Desmond, Lui, Sinclair, Vivash, Willard, O'Brien, Winston, Yasuda, Cendes, Shinohara, Duncan, Iglesias, Adler, Wagstyl.
Drafting of the manuscript: Ripart, Spitzer, Williams, Foldes, Zhang, Lenge, Jackson, Adler, Wagstyl.
Critical review of the manuscript for important

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

intellectual content: Ripart, Spitzer, Fawaz, Williams, Walger, Chen, De Benedictis, De Palma, Napolitano, Rossi-Espagnet, Foldes, Hu, Mo, Likeman, Rüber, Caligiuri, Gambardella, Labate, Güttler, Tietze, Barba, Guerrini, You, Cohen, Gaillard, Tang, S. Wang, I. Wang, Kloster, Pinborg, Davies, Hamandi, Semmelroch, Jackson, Tortora, Severino, Striano, Chari, D'arco, Mankad, Tisdall, Bargalló, Conde-Blanco, Pascual-Diaz, Pariente, Delgado-Martinez, Pérez-Enríquez, González-Ortiz, Abela, Mullatti, Vecchiato, O'Muircheartaigh, Liu, Kalviainen, Kandasamy, McLellan, Sokol, Shetty, Desmond, Lui, Sinclair, Vivash, Willard, O'Brien, Winston, Yasuda, Cendes, Whitaker, Shinohara, Duncan, Cross, Baldeweg, Robinson, Iglesias, Adler, Wagstyl.
Statistical analysis: Ripart, Spitzer, Fawaz, Chen, Napolitano, Foldes, D'arco, Shinohara, Adler, Wagstyl.
Obtained funding: Guerrini, Pinborg, Kalviainen, O'Brien, Winston, Baldeweg, Adler, Wagstyl.
Administrative, technical, or material support: Ripart, Walger, Napolitano, Zhang, Hu, Mo, Caligiuri, Tietze, Lenge, Gaillard, Tang, S. Wang, I. Wang, Hamandi, Semmelroch, Severino, Mankad, Tisdall, Pascual-Diaz, González-Ortiz, Mullatti, O'Muircheartaigh, Kalviainen, McLellan, Yasuda, Cendes, Whitaker, Baldeweg, Robinson, Adler, Wagstyl.
Supervision: Spitzer, De Benedictis, Rüber, Labate, Gaillard, Hamandi, Striano, D'arco, Mankad, Tisdall, Conde-Blanco, Desmond, Whitaker, Shinohara, Duncan, Cross, Baldeweg, Adler, Wagstyl.
Other - image analysis: D'arco.
Other - providing data for the study: Severino.
Other: Gambardella, Kloster, Chari, Pérez-Enríquez.

Conflict of Interest Disclosures: Dr Cohen reported grants from the Pediatric Epilepsy Research Foundation/Child Neurology Foundation Shields Research Grant, Children's National Research Institute Chief Research Officer Award, the American Academy of Neurology Career Development Award, Hess Foundation Research Funds, and support for data acquisition/analysis from District of Columbia Intellectual and Developmental Disabilities Research Center during the conduct of the study. Dr I. Wang reported grants from National Institute of Neurological Disorders and Stroke (R01 NS109439) during the conduct of the study. Dr Kalviainen reported grants from the Saastamoinen Foundation, the Academy of Finland, and Government Funding during the conduct of the study; personal fees from Angelini Pharma, Orion, Eisai, UCB Pharma, Jazz Pharmaceuticals, and grants from the Vaajasalo Foundation outside the submitted work. Dr Vivash reported grants from Eisai and personal fees from Phillips outside the submitted work. Dr O'Brien reported grants from the National Health and Medical Research Council Investigator Grant during the conduct of the study; consulting contracts from UCB Pharma, Jazz Pharmaceuticals, LivaNova, Kinoxis Pharmeceutical, Eisai Pharmaceuticals, and ES Theraputics Research outside the submitted work. Dr Winston reported grants from the Medical Research Council (G0802012, MR/M00841X/1) during the conduct of the study. Dr Cendes reported grants from Sao Paulo Research Foundation (2013075593) during the conduct of the study and honoraria for lectures and advisory board from UCB, Eurofarma-Brazil, Libbs, and Takeda Honoraria outside the submitted work. Dr Shinohara reported grants from the National Institutes of Health during the conduct of the study and personal fees from Octave Bioscience and the American Medical Association outside the submitted work. Dr Wagstyl reported grants from Wellcome Trust, Rosetrees Trust, and the Epilepsy Research Institute during the conduct of the study. No other disclosures were reported.

Funding/Support: Dr Ripart and Dr Adler are supported by the Rosetrees Trust (A2665), Epilepsy Research Institute (P2208), and the National Institute for Health and Care Research Great Ormond Street Hospital Biomedical Research Centre. The views expressed are those of the authors and not necessarily those of the National Health Service, the National Institute for Health and Care Research, or the Department of Health. Dr Napolitano was funded by PCDH19 Alliance. The work of Drs Guerrini, Barba, and Lenge from the Meyer Children's Hospital IRCCS was supported, in part, by funds from the Current Research Annual Funding of the Italian Ministry of Health. Drs Guttler and Tietze were funded by the German Research Foundation (DFG, SFB295RETUNE). Dr Wang was funded by the National Institutes of Health (R01 NS109439). Dr Cohen was supported by a Child Neurology Foundation/Pediatric Epilepsy Research FoundationShields Award, the Children's National Research Institute Chief Research Officer Award, Hess Foundation, and Children's National IDDRC. Dr Pinborg was funded by the Lundbeck Foundation BrainDrugs (R279-2018-1145). The Florey Institute of Neuroscience and Mental Health acknowledges the strong support from the Victorian Government and in particular the funding from the Operational Infrastructure Support Grant. The authors from the Florey Institute of Neuroscience and Mental Health acknowledge the facilities and scientific and technical assistance of the National Imaging Facility, a National Collaborative Research Infrastructure Strategy capability. Drs Liu and Kalviainen acknowledge the Saastamoinen Foundation and Academy of Finland. Dr Shetty was funded by National Health Service Research Scotland. Dr Winston was funded by the Medical Reserve Corp (G0802012, MR/M00841X/1). Dr Yasuda acknowledges support from The National Council of Scientific and Technological Development (315953/2021-7) and São Paulo Research Foundation (2013/07559-3). Dr Duncan was funded by the National Institute for Health and Care Research. Dr Iglesias was funded by the National Institutes of Health (1RF1MH123195, 1R01AG070988, 1R01EB031114, 1UM1MH130981, 1RF1AG080371). Dr Wagstyl was funded by the Wellcome Trust.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

Additional Information: In this study, we developed a graph-based implementation of the nnU-Net model. The nnU-Net model is a convolutional neural network optimized for biomedical segmentation and typically applied on 2D or 3D images. In our graph implementation, the model has been adapted to segment epilepsy lesions from cortical meshes.

REFERENCES

1. Blumcke I, Spreafico R, Haaker G, et al; EEBB Consortium. Histopathological findings in brain tissue obtained during epilepsy surgery. *N Engl J Med*. 2017;377(17):1648-1656. doi:10.1056/NEJMoa1703784

2. Lamberink HJ, Otte WM, Blümcke I, Braun KPJ; European Epilepsy Brain Bank writing group; study group; European Reference Network EpiCARE. Seizure outcome and use of antiepileptic drugs after epilepsy surgery according to histopathological diagnosis: a retrospective multicentre cohort study. *Lancet Neurol*. 2020;19(9):748-757. doi:10.1016/S1474-4422(20)30220-9

3. Téllez-Zenteno JF, Hernández Ronquillo L, Moien-Afshari F, Wiebe S. Surgical outcomes in lesional and non-lesional epilepsy: a systematic review and meta-analysis. *Epilepsy Res*. 2010;89(2-3):310-318. doi:10.1016/j.eplepsyres.2010.02.007

4. Wagstyl K, Whitaker K, Raznahan A, et al. Atlas of lesion locations and postsurgical seizure freedom in focal cortical dysplasia: a MELD study. *Epilepsia*. Published online November 29, 2021. doi:10.1111/epi.17130

5. Spitzer H, Ripart M, Whitaker K, et al. Interpretable surface-based detection of focal cortical dysplasias: a multi-centre epilepsy lesion detection study. *Brain*. 2022;145(11):3859-3871. doi:10.1093/brain/awac224

6. David B, Kröll-Seger J, Schuch F, et al. External validation of automated focal cortical dysplasia detection using morphometric analysis. *Epilepsia*. 2021;62(4):1005-1021. doi:10.1111/epi.16853

7. Gill RS, Lee HM, Caldairou B, et al. Multicenter validation of a deep learning detection algorithm for focal cortical dysplasia. *Neurology*. 2021;97(16):e1571-e1582. doi:10.1212/WNL.0000000000012698

8. Walger L, Adler S, Wagstyl K, et al. Artificial intelligence for the detection of focal cortical dysplasia: challenges in translating algorithms into clinical practice. *Epilepsia*. 2023;64(5):1093-1112. doi:10.1111/epi.17522

9. Cucurull G, Wagstyl K, Casanova A, et al. Convolutional neural networks for mesh-based parcellation of the cerebral cortex. Accessed January 17, 2025. https://openreview.net/pdf?id=rkKvBAiiz

10. Gong S, Chen L, Bronstein M, Zafeiriou S. SpiralNet++: a fast and highly efficient mesh convolution operator. Accessed January 17, 2025. http://arxiv.org/abs/1911.05856

11. Spitzer H, Ripart M, Fawaz A, et al. Robust and generalisable segmentation of subtle epilepsy-causing lesions: a graph convolutional approach. *Medical Image Computing and Computer Assisted Intervention–MICCAI 2023*. 2023:420-428. doi:10.1007/978-3-031-43993-3_41

12. Fawaz A, Williams LZJ, Alansary A, et al. Benchmarking geometric deep learning for cortical segmentation and neurodevelopmental phenotype prediction. *bioRxiv*. Published online December 2, 2021. doi:10.1101/2021.12.01.470730

13. Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nat Methods*. 2021;18(2):203-211. doi:10.1038/s41592-020-01008-z

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025

14. Bronstein MM, Bruna J, LeCun Y, Szlam A, Vandergheynst P. Geometric deep learning: Going beyond Euclidean data. *IEEE Signal Process Mag*. 2017;34(4):18-42. doi:10.1109/MSP.2017.2693418

15. Fischl B. FreeSurfer. *Neuroimage*. 2012;62(2): 774-781. doi:10.1016/j.neuroimage.2012.01.021

16. Adler S, Whitaker K, Semmelroch M, Wagstyl K. MELD protocol 4—lesion masking v2. Accessed January 17, 2025. doi:10.17504/protocols.io.n9udh6w

17. Greve DN, Van der Haegen L, Cai Q, et al. A surface-based analysis of language lateralization and cortical asymmetry. *J Cogn Neurosci*. 2013;25 (9):1477-1492. doi:10.1162/jocn_a_00405

18. Spitzer H, Ripart M, Whitaker K, et al. Interpretable surface-based detection of focal cortical dysplasias: a MELD study. *Brain*. 2022;145 (11):3859–3871. doi:10.1093/brain/awac224

19. Sigalovsky IS, Fischl B, Melcher JR. Mapping an intrinsic MR property of gray matter in auditory cortex of living humans: a possible marker for primary cortex and hemispheric differences. *Neuroimage*. 2006;32(4):1524-1537. doi:10.1016/j.neuroimage.2006.05.023

20. Chen AA, Luo C, Chen Y, Shinohara RT, Shou H; Alzheimer's Disease Neuroimaging Initiative. Privacy-preserving harmonization via distributed ComBat. *Neuroimage*. 2022;248:118822. doi:10.1016/j.neuroimage.2021.118822

21. Ronneberger O, Fischer P, Brox T. *U-Net: Convolutional Networks for Biomedical Image Segmentation*. In: Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015. Springer International Publishing; 2015:234-241.

22. Sundararajan M, Taly A, Yan Q. Axiomatic attribution for deep networks. arXiv [csLG]. Accessed January 30, 2025. https://arxiv.org/pdf/1703.01365

23. MELD Project. MELD classifier using geometric deep learning. Accessed January 22, 2025. https://github.com/MELDProject/meld_graph

24. MELD Project. Prediction of outcome and lesion location. Accessed January 22, 2025. https://github.com/MELDProject/pool

25. Walger L, Bauer T, Kügler D, et al. A quantitative comparison between human and artificial intelligence in the detection of focal cortical dysplasia. *Investigative Radiology*. Published online January 17, 2024. doi:10.1097/RLI.0000000000001125

26. Wang ZI, Alexopoulos AV, Jones SE, Jaisani Z, Najm IM, Prayson RA. The pathology of magnetic-resonance-imaging-negative epilepsy. *Mod Pathol*. 2013;26(8):1051-1058. doi:10.1038/modpathol.2013.52

27. Eriksson MH, Whitaker KJ, Booth J, et al. Pediatric epilepsy surgery from 2000 to 2018: changes in referral and surgical volumes, patient characteristics, genetic testing, and postsurgical outcomes. *Epilepsia*. 2023;64(9):2260-2273. doi:10.1111/epi.17670

28. Czarnetzki C, Spinelli L, Huppertz HJ, et al. Yield of non-invasive imaging in MRI-negative focal epilepsy. *J Neurol*. 2024;271(2):995-1003. doi:10.1007/s00415-023-11987-6

29. McGonigal A, Bartolomei F, Régis J, et al. Stereoelectroencephalography in presurgical assessment of MRI-negative epilepsy. *Brain*. 2007; 130(Pt 12):3169-3183. doi:10.1093/brain/awm218

30. Eriksson MH, Prentice F, Piper RJ, et al. Long-term neuropsychological trajectories in children with epilepsy: does surgery halt decline? *Brain*. 2024;147(8):2791-2802. doi:10.1093/brain/awae121

31. Hale AT, Chari A, Scott RC, et al. Expedited epilepsy surgery prior to drug resistance in children: a frontier worth crossing? *Brain*. 2022;145(11): 3755-3762. doi:10.1093/brain/awac275

32. Adler S, Wagstyl K, Gunny R, et al. Novel surface features for automated detection of focal cortical dysplasias in paediatric epilepsy. *Neuroimage Clin*. 2016;14:18-27. doi:10.1016/j.nicl.2016.12.030

33. Jin B, Krishnan B, Adler S, et al; Pediatric Imaging, Neurocognition, and Genetics Study. Automated detection of focal cortical dysplasia type II with surface-based magnetic resonance imaging postprocessing and machine learning. *Epilepsia*. 2018;59(5):982-992. doi:10.1111/epi.14064

34. Wagstyl K, Adler S, Pimpel B, et al. Planning stereoelectroencephalography using automated lesion detection: retrospective feasibility study. *Epilepsia*. 2020;61(7):1406-1416. doi:10.1111/epi.16574

35. Chari A, Adler S, Wagstyl K, et al. Lesion detection in epilepsy surgery: lessons from a prospective evaluation of a machine learning algorithm. *Dev Med Child Neurol*. 2024;66(2): 216-225. doi:10.1111/dmcn.15727

36. Naeini MP, Cooper G, Hauskrecht M. Obtaining well calibrated probabilities using bayesian binning. *AAAI*. 2015;29(1). doi:10.1609/aaai.v29i1.9602

37. Urbach H, Kellner E, Kremers N, Blümcke I, Demerath T. MRI of focal cortical dysplasia. *Neuroradiology*. 2022;64(3):443-452. doi:10.1007/s00234-021-02865-x

38. Duncan JS. Multidisciplinary team meetings: the epilepsy experience. *Pract Neurol*. 2022; practneurol-2022-003350. doi:10.1136/practneurol-2022-003350

39. Baumgartner C, Koren JP, Britto-Arias M, Zoche L, Pirker S. Presurgical epilepsy evaluation and epilepsy surgery. F1000Res. 2019;8. doi:10.12688/f1000research.17714.1

40. Cohen NT, You X, Krishnamurthy M, et al; Multi-Centre Epilepsy Lesion Detection (MELD) Project. Networks underlie temporal onset of dysplasia-related epilepsy: a MELD study. *Ann Neurol*. 2022;92(3):503-511. doi:10.1002/ana.26442

41. Mellor S, Timms RC, O'Neill GC, et al; MELD Project Consortium. Combining OPM and lesion mapping data for epilepsy surgery planning: a simulation study. *Sci Rep*. 2024;14(1):2882. doi:10.1038/s41598-024-51857-3

© 2025 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by University of Rochester user on 09/05/2025